



Chancellor College

Department of Computer Science

Second Semester Test Two Examinations

Examination Code: **INF221**

Examination Title: **Web Design and Development**

Duration: **Two Hours**

Date: **20th December 2020**

Time: **2:00pm – 4:00pm**

Instructions

- a) This paper contains **three** pages (including this cover page), please check.
- b) This paper contains **FOUR** questions. You are required to answer **all** the questions.
- c) The maximum **marks** attainable from each question are shown in brackets to the right of each question.
- d) Show your reasoning. Unsupported answers will lead to loss of marks.

Pass Rate: 36%

Highest: 80% (Ng'oma, C. Mr.; Both Test 1 & 2)

Lowest: 19%

Avg: 44%

Question 1 - General (21 Marks)

a. Define the following: (1 Marks)

i. Event propagation

- i. describes the "stack" of events that need to fire up in the browser.
- ii. defines the order/manner in which events are executed.
- iii. relates to the order in which event handlers are called when one element is nested inside a second element, and both elements have registered a listener for the same event (onclick, onchange, etc.)
- iv. event bubbling, capturing - *describe what these two terms are*

b. Javascript is considered a multi-paradigm programming language. List and explain any two reasons? (4 marks)

- i. imperative - object oriented programming supported
- ii. declarative - functional programming supported

c. What is the difference between synchronous and asynchronous programming? (4 marks)

- e. Explain any three states defined by the JavaScript ***Promise*** object. (3 marks)
- ii. *pending*: initial state, neither fulfilled nor rejected.
 - iii. *fulfilled*: meaning that the operation was completed successfully.
 - iv. *rejected*: meaning that the operation failed.
- i. Using Javascript, write a method to demonstrate the concept. (2 marks)
- ```
const promise = new Promise((resolve, reject) => { })

promise.then(res => {})

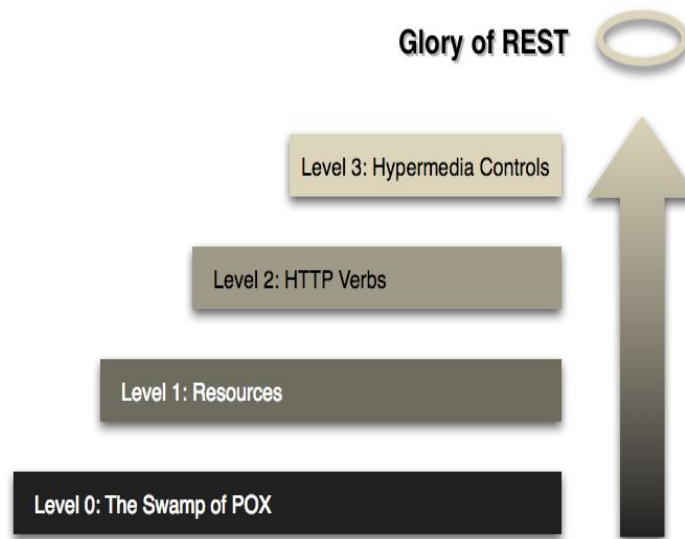
promise.catch(error => console.log(error))
```
- f. Using a well labelled diagram. Discuss how AJAX works in web communications. (6 marks) - [W3Schools or refer to class demo diagrams]
- g. State the default port for the HTTP protocol? (1 mark) – 80

## Question 2 - REST (34 Marks)

- b. What does the following stand for: (2 mark)
- i. REST - REpresentational State Transfer
  - ii. HATEOAS – HyperText As The Engine of Application State
- c. Explain any five methods/verbs defined by the HTTP protocol. (5 marks)
- i. GET, POST, PUT, DELETE, PATCH, HEAD, OPTIONS etc
- d. “The ***REST architectural style*** includes the ***design constraints*** which have been followed to define the HTTP protocol,...the ***fundamental standard*** together with ***URI*** (scheme://host[:port#]/path/.../[,url-params][?query-string][#anchor]) and ***HTML*** which has enabled to build the Web” C. Pautasso (2014)
- i. Explain any three non-optional parts of a URI – for HTTP or HTTPs. (3 marks)

- i. Scheme - identifies the protocol used to access the resource on the Internet. Can be HTTP (without SSL) or HTTPS (with SSL)
  - ii. Host - identifies where the resource is held, the host that holds the resource
  - iii. Path - identifies the specific resource in the host that the web client wants to access.
- ii. List any four benefits of REST architectural style. **(4 marks)**
- i. Simplicity
  - ii. Scalability
  - iii. Portability
  - iv. Visibility
  - v. Reliability
  - vi. Separation of concerns
  - vii. Performance
- iii. Discuss the three design constraints of the REST architectural style and how each impacts the design of **TRUE** RESTful Web services. **(8 marks)**
- i. Addressability - unique & stable resource identifiers
  - ii. Uniform Interface - resource interaction is through a uniform interaction provided by GET, POST, PATCH, DELETE, OPTIONS, HEAD
  - iii. Stateless Interactions - client/server permanent sessions are not established to span multiple interactions
  - iv. Self-Describing Messages - responses and requests contain both data and meta-data (description/information about data)
  - v. Hypermedia - relationship between resources

- iv. Using a well labelled diagram, discuss the four levels of maturity of the API services that use REST architectural style. (8 marks)



**Level 0:** HTTP as a tunnel These are all services which simply exchange XML documents (sometimes referred to as Plain-Old-XML documents as opposed to SOAP messages) over HTTP POST request and responses, effectively following somekindofXML-RPC protocol.

**Level 1:** As opposed to using a single endpoint for tunneling RPC messages through the HTTP protocol, services on maturity level 1 make use of multiple identifiers to distinguish different resources. Each interaction is addressed to a specific resource.

**Level 2:** make proper use of the REST uniform interface in general and of the HTTP verbs in particular.

**Level 3:** These are the fully mature RESTful Web services, which in addition to exposing multiple addressable

resources which share the same uniform interface also make use of hypermedia to model relationships between resources. This is achieved by embedding so-called hypermedia controls within resource representations

e. Discuss any major difference(s) between RESTful Web services and WS-\* Web services with respect to **resources** and their **representations**. (4 marks, Free Marks, no one got it right)

i. RESTful

- i. resources can be accessed via multiple (URIs) communication endpoints
- ii. resources represented using multiple message formats (XML, JSON, ATOM, HTML, CSV, etc.)

ii. WS-\*

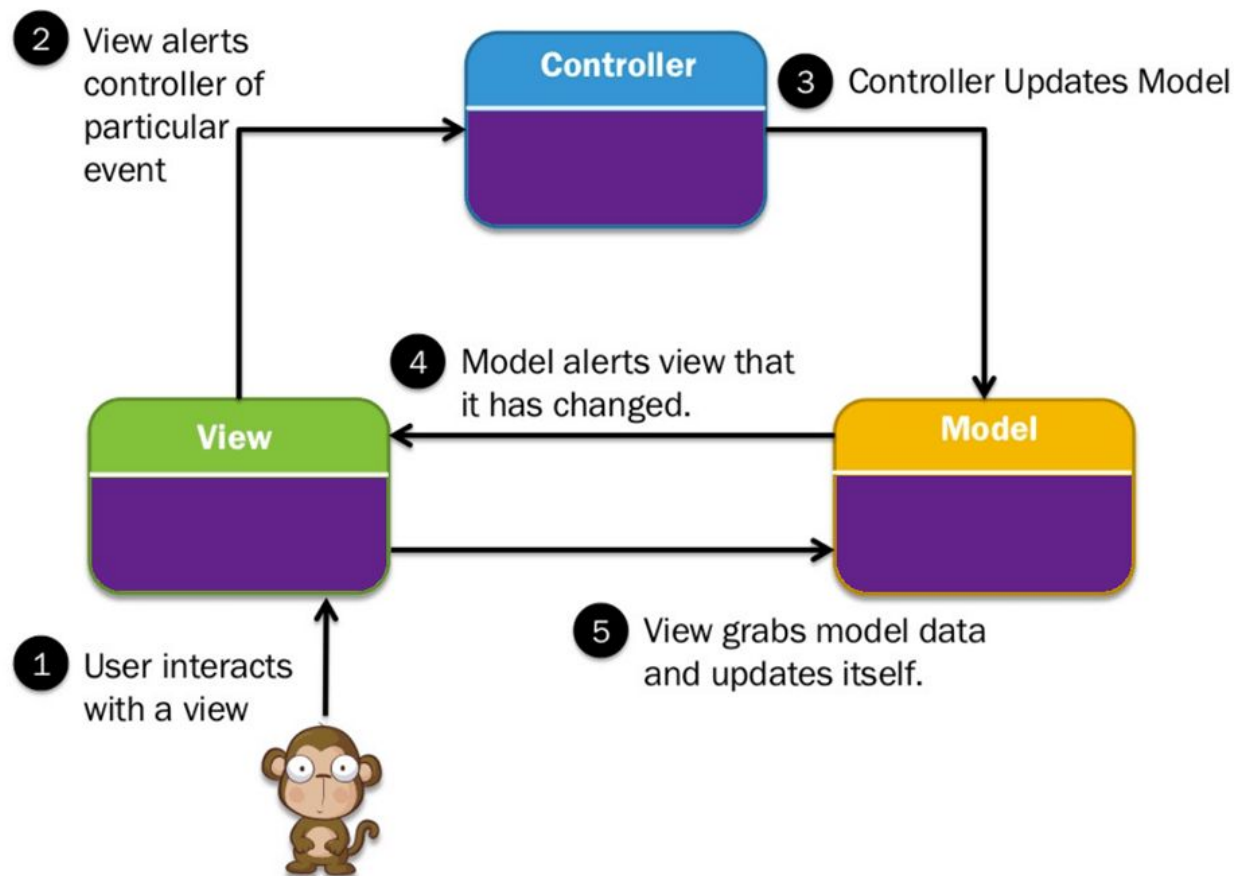
- i. resources can be accessed via a single communication endpoint - Remote Procedure Call (RPC)
- ii. resources represented using a single message format (SOAP)

### Question 3 - Web Frameworks (29 Marks)

a. List any two software architectural patterns beside MVC. (2 marks)

- i. MVP - Model View Presenter
- ii. MVVM - Model View ViewModel or Model View Binder

b. With a well labelled diagram explain how different components in the MVC software architectural pattern work together. (6 marks)



c. Define the following term: **(3 marks)**

a. Software Framework

i. software framework is an abstraction in which software providing generic functionality can be selectively changed by additional user-written code, thus providing application-specific software. A software framework provides a standard way to build and deploy applications

d. Explain any two advantages of using UI frameworks in software development. **(4 marks)**

- i. Investing in the Task, Not in the Technology
- ii. Guaranteed Upgradability & Maintenance
- iii. Efficiency
- iv. Integration

- v. Tooling
  - vi. Compartmentalization
  - vii. Routing
- e. List any two Web UI frameworks, based on Javascript, beside ReactJS. **(2 marks)**
- i. Vue
  - ii. Angular
  - iii. Ember
  - iv. etc
- f. Explain any theoretical concept(s) defining any of the frameworks you listed above, including ReactJS. **(6 marks)**
- i. data binding
  - ii. virtual dom
  - iii. jsx
  - iv. etc.
- g. Explain why JSX is important as a Javascript template language for React. Give any three reasons. **(6 marks)**
- i. Binding content
  - ii. Binding properties - Specifying Attributes, Specifying Children
  - iii. Binding events

#### **Question 4 - Principles, Patterns & Web Processes (16 Marks)**

- a. Explain any two characteristics of good software. **(4 marks)**
- i. Maintainable
  - ii. Testable
- b. List and explain any two key phases of the software development process. **(4 marks)**
- i. Research & Planning or Requirements Gathering
  - ii. Design or Prototype
  - iii. Develop

- iv. Testing - can be marked
- v. Deployment & Maintenance - can be marked
- c. Discuss any two reasons why understanding organization's business processes helps in defining the software project goals. **(4 marks)**
  - i. [Application here]
- d. Give any two reasons why writing software tests is important in software development. **(4 marks)**
  - i. Validation - if what is being developed or has been developed really meets the business requirements. Or processes the inputs and gives the right expected outputs.
  - ii.